CS 367 - DPLL Handout

November 4, 2026

Trace the DPLL algorithm on the following boolean formulae:

$$F = (A \vee \neg B) \wedge (\neg A \vee C) \wedge (B \vee \neg C)$$

$$F = (A) \wedge (B) \wedge (\neg A \vee C) \wedge (\neg B \vee \neg C)$$

1 Step 1:

Boolean Constant Propagation. If any clause has a single literal (called a unit clause), set the literal to True (or False if it is negated!) and simplify the formula. When no simplification can occur, skip to step 2

Iter Num: _____
Simplified Formula: _____

Iter Num: _____
Simplified Formula: _____

Iter Num: _____
Simplified Formula: _____

Iter Num: _____
Simplified Formula: _____

2 Step 2:

Search. Choose a literal (e.g., A) and assign it a value (True or False). What is the simplified formula after this step?

Iter Num: _____
Literal and Value Chosen: _____
Simplified Formula: _____

Iter Num: _____
Literal and Value chosen: _____
Simplified Formula: _____

Recursively trace DPLL on the simplified formula

3 Termination

Report if the formulae were SAT or UNSAT. If satisfiable, provide an interpretation.